

Unraveling the Folding Dynamics of DNA Origami Structures

Meysam Mohammadi-Zerankeshi, James Houston, Ogochukwu K.U. Elisha-Wigwe, Abi Sachi, and Alexander E. Marras*

Achieving high folding yield remains a challenge in DNA origami, particularly as structures increase in complexity and scale. Here, how DNA origami design influences folding is investigated using a combination of real-time fluorometry, gel electrophoresis, electron microscopy, and theoretical analysis. Results reveal a balance of free energy changes from loop formation and hybridization that govern nucleation of nanostructure assembly, while the extent of cooperativity determines the overall assembly. The effect of structural complexity, staple design, and scaffold design on each energetic parameter, folding yield, kinetics, and cooperativity is measured. The results show that the scaffold pattern determines the extent of cooperativity, where fewer scaffold crossovers result in more cooperative folding. These findings use a tool developed in this work to estimate the extent of cooperativity in any structure. It is also found that limiting the number of crossovers per staple should be prioritized over extending staple binding domains, as the entropic penalty dominates the favorable binding. Finally, a 1–2 h focused annealing ramp strategy is demonstrated, that can increase yield up to 17% relative to traditional multi-day ramps. Optimizing energy changes and cooperativity through design can significantly enhance assembly yield and reduce time, particularly for complex structures, aiding large-scale DNA materials.

1. Introduction

The development of DNA-based nanomaterials through the controlled self-assembly of engineered nucleotide sequences is responsible for the best geometric precision of any soft material. Recent advances using DNA origami include shells for trapping viruses,^[1] a ratchet-driven molecular rotor,^[2] and a tunable molecular force sensor.^[3] Further assembly efforts expand

DNA origami technology to larger scales through hierarchical assembly of nanostructures,^[4–7] requiring high yield and larger scale production of nanostructures. However, producing DNA nanostructures rapidly and at high yields are still significant challenges that should be addressed to reach the full potential of DNA origami.^[8–10] A comprehensive understanding of the complex DNA origami assembly process will bring us closer to this goal.

DNA origami is fabricated via a long single-stranded DNA “scaffold” that is folded into predetermined 2D or 3D shapes by many short “staple” DNA oligonucleotides that are the complementary counterparts of the scaffold sequence.^[11,12] The result can be a vast range of nanoscale devices and structures with applications spanning drug carriers,^[13,14] mechanically functional nanomaterials,^[15–17] biosensor platforms,^[18–20] for arrangement of molecules,^[21,22] and much more. Unfortunately, with complex designs some applications become limited due to poor fabrication yield and the typical

requirement of a multi-day fabrication process, but recent studies are revealing paths to overcome these barriers. Sobczak et al.^[8] compared the folding behavior of DNA origami objects to reveal the significant role of staple routing and cooperativity. By employing constant temperature annealing, they enhanced the folding yield and sped up the fabrication time. Martin and Dietz^[17] showed that emphasizing longer continuous staple domains improved folding yield. Recently Aksel et al.^[9] introduced a computational framework to optimize staple routing based on robust thermodynamic models to increase yield and precision of DNA origami fabrication. In this study, machine learning enabled auto-breaking of staples of any given design that would result in the highest yield. Furthermore, DeLuca et al.^[23] employed a mesoscopic simulation model that revealed a multi-step folding process for DNA origami, where structures initially zip into a partially folded precursor before crystallizing into their final form. In another study,^[24] Monte Carlo simulations showed that nucleation barriers of staple binding are design-dependent, with coaxial stacking of adjacent staples playing a critical role in assembly time and yield. Even with advancements in understanding the DNA origami folding process, the intricate relationship between nanostructure design and assembly behavior for complex

M. Mohammadi-Zerankeshi, J. Houston, O. K. Elisha-Wigwe, A. Sachi, A. E. Marras
Walker Department of Mechanical Engineering
University of Texas at Austin
Austin, TX 78712, USA
E-mail: amarras@utexas.edu

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/smll.202504855>

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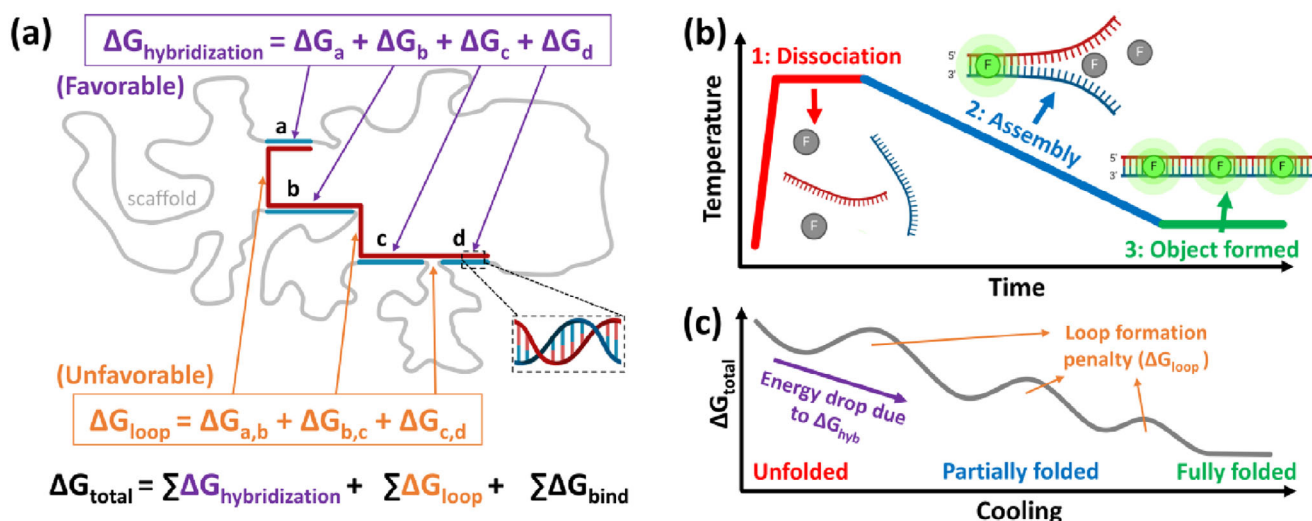


Figure 1. Free energy changes in DNA origami folding. a) Total free energy changes during the assembly process as the result of the interactions between staples and sections of scaffold by favorable hybridization (ΔG_{hyb}), unfavorable entropic cost from loop formation (ΔG_{loop}), and binding of staples (ΔG_{bind}). The schematic depicts the hybridization of one staple to different sections of the scaffold labeled (a–d), inspired by the model from Aksel et al.^[9] b) Characterizing the assembly of DNA origami by real-time fluorimetry involves 1) heating to dissociate all base pairs, 2) slowly cooling to allow for equilibrium assembly, resulting in 3) the final object properly formed. The fluorescence agent (F) attaches and fluoresces only to double-stranded DNA, signifying formation. c) Total free energy changes in the system during assembly, where the fully assembled structure has a lower total free energy than the unpaired scaffold and staple DNA as the result of hybridization. The system should overcome loop formation penalty barriers during assembly, wherein the degree of such increase in energy would be lower in later peaks due to the cooperativity effect.

structures remains vague. Many parameters including scaffold routing, staple break strategies, and folding protocols influence assembly through changes in local and global free energy and the role of cooperativity. Further focused studies are required to fully characterize this process.

There are three primary thermodynamic factors that regulate the binding of a staple to the associated section of the scaffold, and hence, the total assembly behavior of the nanostructure,^[25–27] as recently described in Aksel et al.^[9] and depicted in **Figure 1a**. ΔG_{hyb} is the free energy change due to scaffold–staple hybridization, which is a favorable term because the formed duplex has lower energy than the single-stranded counterparts. In this computational model,^[9] ΔG_{hyb} was determined by calculating the enthalpic gain from base pairing between a staple and the scaffold, using the nearest-neighbor thermodynamic model,^[26] which accounts for sequence-specific base stacking interactions. In this model, free energy calculations were carried out at 50 °C, a temperature at which nearest-neighbor thermodynamic predictions are most reliable and which closely aligns with conditions commonly used in experimental DNA origami folding. The second term, ΔG_{loop} , is the energy cost to the system from decreasing entropy by folding the scaffold and overcoming electrostatic repulsion, an unfavorable energy change. To calculate ΔG_{loop} , the scaffold was modeled as a freely jointed chain,^[25] and the average end-to-end distance between the looped segments was used to estimate an effective local concentration of one segment near the other. Longer loops result in lower such concentration values and thus larger entropic penalties. The third term is the energy required for the diffusion of the corresponding staple, shown by ΔG_{bind} . The ΔG_{bind} is a concentration-dependent term that reflects the entropic cost of bringing a staple from solution into proximity with its complementary site on the scaffold, a bimolec-

ular association step.^[25] Continuing to follow the methodology used in Aksel et al.,^[9] this term was treated as a constant of +10.3 kcal mol^{−1} across all staples and designs (Equation S1, Supporting Information). These energy changes collectively contribute to the total free energy change (ΔG_{total}) of the staple during binding, playing a crucial role in the assembly process.^[28,29]

However, these thermodynamic terms are not the only parameters that govern the folding behavior of DNA origami structures. Namely, DNA origami folding is a highly cooperative process, often with a nucleation barrier. During folding, the lowest energy staple binding domains should bind first, mainly determined by length and sequence (e.g., ΔG_b in Figure 1a). Once a second binding domain from the same staple is bound (e.g., ΔG_a and ΔG_b), closing a scaffold loop, the adjacent staple bridging the same two scaffold sections will have a facilitated binding event due to the reduced entropy penalty (e.g., $\Delta G_{a,b}$) necessary for binding. This event will facilitate cooperative assembly of the remaining staples in this region, and in some cases, throughout the entire structure. For many DNA origami designs, a critical number of initial binding events must occur before the structure can proceed efficiently toward its fully folded state, a process limited by what is known as the nucleation barrier.^[30] When adjacent staples on the same helix coaxially stack or the number of staple connections between scaffold sections increases, they create strong cooperative behavior. This cooperativity promotes multiple staple bindings to occur together, helping the structure overcome the nucleation barrier and stabilize partially folded intermediates, facilitating the full assembly.^[24,30] Contrary to folding, unfolding during the melting process is mostly independent of neighboring staple interactions, and usually happens at higher temperatures than the temperature at which the structure fully assembles, resulting in a hysteresis dependent on design complexity.^[30]

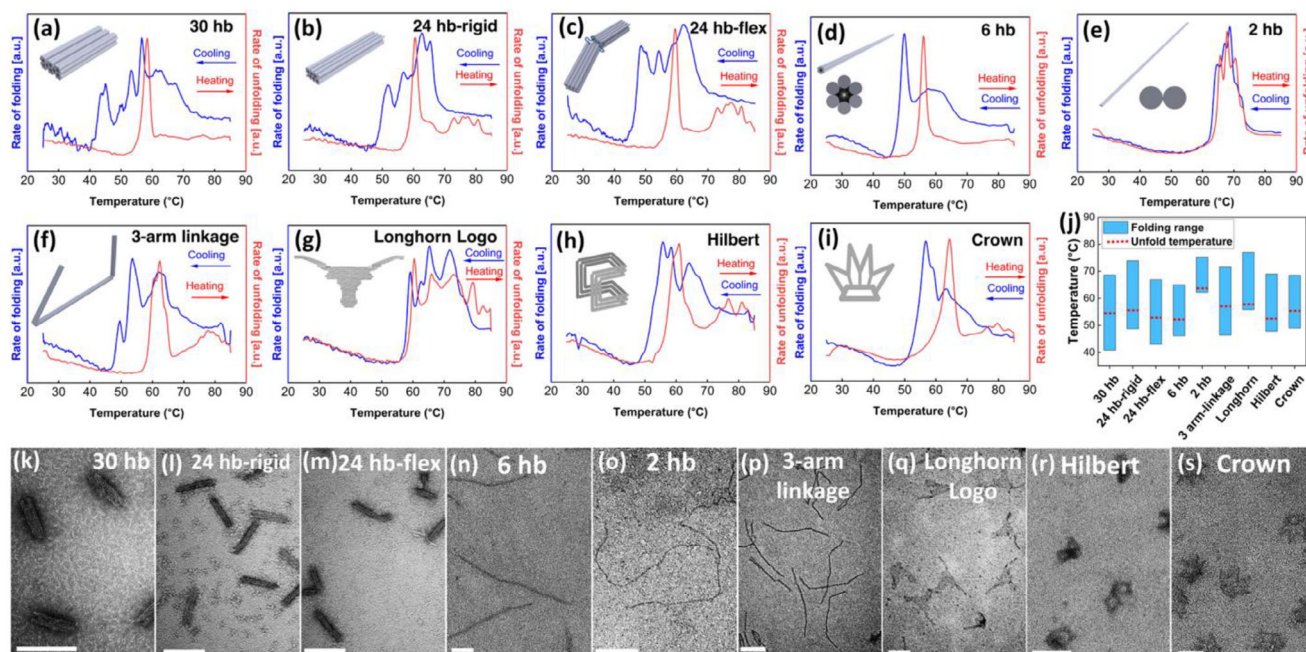


Figure 2. Folding and unfolding real-time fluorometry results of a) 30 hb, b) 24 hb-rigid, c) 24 hb-flex, d) 6 hb, e) 2 hb, f) 3-arm linkage, g) Longhorn logo, h) Hilbert, and i) Crown structures. Rates of folding and unfolding calculated from the change in fluorescence (hybridization) at each temperature. j) Folding range and start of unfolding temperatures from a–i). TEM images of folded structures: k) 30 hb, l) 24 hb-rigid, m) 24 hb-flex, n) 6 hb, o) 2 hb, p) 3-arm linkage, q) Longhorn logo, r) Hilbert and s) Crown structures. Scale bars = 70 nm.

The cumulative role of individual scaffold-staple interactions does not fully describe the resulting nanostructure folding behavior. Instead, folding is primarily driven by cooperative effects due to the successive reduction in entropic penalty as scaffold loops form. Models like Aksel et al.⁹ that do not consider cooperativity fall short in their holistic model of folding behavior. Some computational efforts have introduced cooperative effects via local mechanisms such as loop entropy, coaxial stacking, and conditional neighborhood interactions.^[30–32] However, these models treat cooperativity primarily as a local, pairwise phenomenon and therefore fail to fully capture the design-dependent cooperativity observed in folding experiments. This study analyzes the folding behavior of a diverse set of DNA origami structures to elucidate the impact of geometric complexity and staple design on cooperativity and folding behavior. We explore the relationship between design and the extent of cooperativity in assembly and examine the impact on assembly yield and time. The structures include a 30 helix-bundle (hb), flexible and rigid 24 hbs, a 6 hb, 2 hb, 3 arm-linkage, flat single-layer Longhorn logo, and two complex designs from the Castro group's previously published MagicDNA 2.0 work,^[33] the Hilbert and Crown. In addition, scaffold design strategies for the same structure are compared. Experimental folding characterization is performed through gel electrophoresis, electron microscopy, and real-time fluorometry during thermal annealing to analyze the folding across various time points and temperatures. The computational platform recently introduced by Aksel et al.^[9] is implemented to calculate theoretical energy profiles and compared to experimental results. Finally, using a computational tool introduced in this research, one can calculate our estimate of the extent of cooperativity in DNA origami designs using inputs from real-time fluorometry data

and folding energies calculated by the computational model in the Aksel et al.^[9] work.

2. Results and Discussion

2.1. Folding and Unfolding Behavior

Real-time fluorometry experiments revealed distinct folding patterns for all DNA origami structures. Each initiates, progresses, and completes assembly at unique temperatures, offering insights into the assembly process. DNA origami nanostructures are typically assembled by heating the components above the melting temperature of all staples, then gradually cooling the solution to properly anneal each staple to the respective scaffold sections to form the intended structure. In this work, we perform these annealing ramps via real-time fluorometry with SYBR Green dye added in solution. The dye will intercalate and start to fluoresce only once DNA hybridizes, providing a real-time read-out of DNA origami folding (Figure 1b), as first demonstrated by Sobczak et al.^[8] If heated again, the structure will disassemble, as indicated by a loss of fluorescence. Results from fluorometry experiments in Figure 2a–f show the rate for folding (cooling) and unfolding (heating) at each ramp temperature for all nanostructures studied. This includes a 30 hb, 24 hb, flexible 2-arm version of the 24 hb, 6 hb, 2 hb, 2 hb, 3-arm 8 hb (3 arm-linkage), single-layer 2D Longhorn logo, Hilbert structure, and crown. Each peak indicates a steep transition in folding behavior, calculated from fluorescence intensity changes as the amount of hybridization in solution changes. Folding ranges and the start of rapid unfolding are compared in Figure 2j, showing a narrower folding range at higher temperatures for the simpler 2D and 1D structures.

Table 1. Comparing the theoretical and experimental T_{fold} , experimental T_{unfold} , hysteresis, width of folding range, total number of scaffold crossovers, number of internal scaffold crossovers not bridged by staples, percentage of 14 nt seeds and the contribution of cooperativity to the assembly of each structure.

Structure	Theoretical T_{fold} [°C]	Experimental T_{fold} [°C]	Experimental T_{unfold} [°C]	Hysteresis [$T_{\text{fold}} - T_{\text{unfold}}$] [°C]	Width of folding range [°C]	Total scaffold crossovers	Unbridged scaffold crossovers	Staples 14 nt seed [%]	Role of cooperativity [%]
2 hb	55.2	62.2	63.4	1.6	13.0	2	–	98	98.2
6 hb	39.3	46.1	52.3	6.2	18.9	10	–	91	83.5
24 hb-rigid	43.1	48.7	55.8	6.3	18.4	46	–	85	72.6
Crown	43.2	49.0	55.3	6.3	19.5	56 ^{a)}	–	89	67.5
30 hb	37.0	40.8	54.6	13.8	27.8	58	–	51	61.7
Hilbert	46.9	47.8	52.1	4.3	21.2	46 ^{a)}	–	81	52.7
24 hb-flex	43.0	43.0	53.0	10.0	23.9	46	22	81	51.1
3-arm linkage	44.9	46.4	56.6	10.2	25.2	30	16	71	50.0
Longhorn	61.7	55.8	57.8	2.0	21.2	90	–	95	19.2

^{a)} Rigid connections with no separation but a change in angle (e.g., Crown spikes) were treated as single continuous helices.

Agarose gel electrophoresis (Figure S1, Supporting Information) and transmission electron microscopy (TEM) (Figure 2k–s) confirm nanostructure assembly. Also, TEM images demonstrate that once folded, structures are stable up to their melting temperature (above T_{fold}) and dissociate above their melting temperature (Figures S2 and S3, Supporting Information). The thermal ramp rate was optimized for the proper assembly of all structures (Figures S4–S6, Supporting Information).

A key metric in folding analysis is the temperature at which folding is completed, T_{fold} , which is seen here as the left side of the lowest temperature peak where no noticeable changes in folding are observed at temperatures below T_{fold} .^[8] We observe drastic differences in T_{fold} as well as the number of distinct assembly steps between designs, where structure density and complexity result in lower T_{fold} and more folding steps. This is likely due to the greater entropic cost and embedded helices hindering folding of complex structures. Also, due to the greater number of crossovers per staple in a complex 3D structure, they are more likely to experience kinetic traps that can hinder or slow the folding process.^[34] For instance, in the case of the 30 hb (many densely packed helices), folding starts at ≈ 68 °C and is completed at 41 °C with many folding steps. On the other hand, the 2 hb starts assembling at a higher 74 °C and completes the process at 61 °C, a narrow folding range of only 13 °C. Between these extremes, the 24 hb folds in a moderate 68–48 °C range. Adding complexity to the 24 hb with a single-stranded scaffold joint (24 hb-flex) lowers T_{fold} and adds more distinct folding steps due to the loss of cooperativity between arms, as discussed later. Hilbert and Crown structures constructed from different small sections such as 2, 6 and 8 hb demonstrate T_{folds} at 47.8 and 49.0 °C, respectively. Full results are shown in Figure 2 and Table 1.

In a larger bundle of helices, there are more opportunities in the design to cross staples over to a neighboring helix, leading to a higher crossover density. The higher crossover density in the 30 hb created a nucleation barrier that requires cooperative binding of staples to initiate folding, whereas the 2 hb lacks a significant nucleation barrier, leading to its rapid and stochastic folding at higher temperatures.^[24] The presence of a nucleation barrier lowers the T_{fold} in the 30 hb due to the stronger entropic con-

tribution in the early stages of assembly (higher temperatures). As temperature is lowered during annealing, the reduced thermal fluctuations and decreased entropic cost (Equation 1) enable the system to overcome this barrier and complete folding. Constant temperature annealing of 30 hb at 52, 49 and 41 °C at different time points shows that this structure would fold faster at the lower temperature, likely due to the reduced entropic penalty (Figure S8, Supporting Information).

$$\Delta G_{\text{total}} = \Delta H_{\text{total}} - T \Delta S_{\text{total}} \quad (1)$$

Contrary to the folding process, during unfolding, staple dissociation is not dependent on other staples. This phenomenon is mostly governed by the average penalty in the free energy of the object, leading to a sharp peak in the rate of unfolding, usually with a higher unfolding temperature (T_{unfold}) relative to T_{fold} . Staple binding order does not play a significant role in unfolding,^[30,8] as it does in folding. Once the structure is formed, the average hybridization energy is the same for all structures with similar scaffolds and staple design strategies. The average free energy penalty determines the unfolding process, meaning structures with higher penalty energy will unfold at lower temperatures. T_{fold} , T_{unfold} , and the difference between them are summarized in Figure 2j and Table 1 for each structure. We find that the additional complexity of structures with many helices or disjointed sections result in more hysteresis. Minimal hysteresis was seen in simple designs like the 1D 2 hb and 2D Longhorn structures. The presence of a nucleation barrier in the 3D designs is the primary reason for hysteresis, where unfolding and folding follow different pathways. Complex structures with a higher nucleation barrier (e.g., 30 hb) exhibit a more significant hysteresis than simple designs (e.g., 2 hb) with minimal nucleation barrier. The degree of hysteresis for each structure is given in Table 1.

Differences in structure complexity (e.g., 30 hb vs 2 hb) influence many scaffold-staple design factors. The higher crossover density for structures like the 30 hb means each scaffold-staple pairing domain between crossovers will be shorter, resulting in a higher and less favorable ΔG_{hyb} . More helices and crossovers also means looping more scaffold sections, which increases the

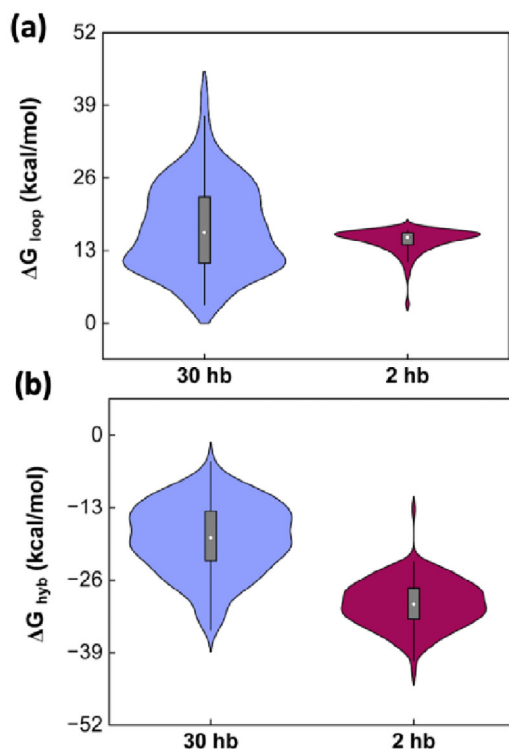


Figure 3. Theoretical energy contributions for every staple in the 30 and 2 hb structures. a) ΔG_{loop} and b) ΔG_{hyb} show narrower energy distributions for 2 hb structure compared to 30 hb due to a more ordered scaffold-staple design, leading to a smaller width of folding range. The dark rectangles represent the 25–75% percentile of the data, the bars show the 1.5 interquartile range, and the circle is the median.

variability in ΔG_{loop} values across the structure due to differences in loop lengths and local geometries. For instance, the 2 hb has potential staple crossovers every 21 bp, compared to every 7 bp in the 30 hb design, resulting in a consistently ordered staple design where almost every staple is nearly identical in length and crossovers, differing only by sequence (Figure S13, Supporting Information). Such a consistent ordered staple design leads to a narrow total free energy change (Figure 3). The 30 hb has far more heterogeneous staple lengths and crossover patterns and thus has greater free energy changes during assembly. The 2 hb also has only 2 scaffold crossovers, compared to 58 on the 30 hb, contributing to the differences in ΔG_{loop} . This result is directly seen in folding and unfolding curves in Figure 2 where the 2 hb has a narrow folding range, 21.4 °C higher T_{fold} , and minimal hysteresis, compared to the 30 hb structure. The loop and hybridization energies for all studied structures are presented in Figure S9 (Supporting Information).

2.2. Impact of Staple Routing on Folding Behavior

The dynamic nature of the free energy of each staple plays an important role in folding behavior. Staple energies change throughout the folding process as the system changes and can be directed through scaffold-staple design.^[35] According to thermodynamic terms discussed in the introduction, increasing the length of staples and decreasing the number of crossovers can lead to a

lower total free energy change, hence facilitating binding events as demonstrated with the results in Figure 3. However, achieving both factors and maintaining the structural integrity of the object is difficult, especially in complex designs. In this regard, to evaluate the influence of staple length and the number of crossovers, staple length was increased for the 3 arm-linkage at the cost of more crossovers per staple, as depicted in Figure 4a,b. In these designs, the circled staple breaks/nicks were removed throughout the structure, resulting in longer staple binding sections and longer overall staples. In total, 37 nicks were removed in the modified design with no other changes. Agarose gel electrophoresis (Figure 4c) and TEM images (Figure S10, Supporting Information) showed that the design with longer staples demonstrated a significant drop in yield, from 94% to 20%, despite the addition of longer binding domains. The free energy changes of loop formation and hybridization of each staple are plotted in Figure 4d,e. Although the longer staple design noticeably lowers ΔG_{hyb} , the same staples resulted in higher ΔG_{loop} due to the five crossovers necessary for proper binding of the highlighted staples, compared to 2–3 crossovers per staple in the original design. This means the long staple design requires the closing of five scaffold loops per staple, leading to a higher entropy penalty. This increases the chance of kinetic traps that hinder consecutive staple pairing and can also reduce the local cooperativity effect. Majikes et al.^[34] also observed that higher loop entropy can increase the chance of a potential blocked state in the self-assembly process. Therefore, we see further evidence that the entropy cost plays the larger role in the assembly behavior compared to the energy of hybridization.

Moreover, when each arm is analyzed individually we see a similar trend where the long staple versions of all three arms have lower ΔG_{total} . For instance, Table S1 (Supporting Information) shows that by increasing the average staple length for Arm 1 from 31.65 to 37.21 and the average number of staple crossovers in that arm by 0.53, the average ΔG_{hyb} is decreased by 4.01 and the average ΔG_{loop} is increased by 2.05 kcal mol^{−1}. This leads to a −1.96 kcal mol^{−1} change in total free energy, which contradicts the decreased yield seen experimentally. We attribute this observation to the higher number of crossovers per staple, which increases the average entropy cost of each arm. Staples with more than three crossovers can cause an over-constraining effect during assembly, which is not well-captured in computational tools. Therefore, our design guidance to enhance yield is to still prioritize long staple binding domains (14+ bp), but to limit the number of crossovers per staple to 3.

2.3. The Role of Cooperativity

Cooperativity is present in both intrastrand (in the subsequence of the same strand) and interstrand (between different strands) folding.^[36] By increasing the length of a staple, intrastrand cooperativity is improved, where initial base pairs help remaining base pairs by reducing their entropic penalty. This is responsible for the enhanced folding yield when including 14 bp seeds, as observed by Martin and Dietz.^[35] However, our results in Figure 4 show that this also leads to a significant increase in entropy cost by increasing the number of crossovers per staple. The increased intrastrand cooperativity is not significant enough to enhance

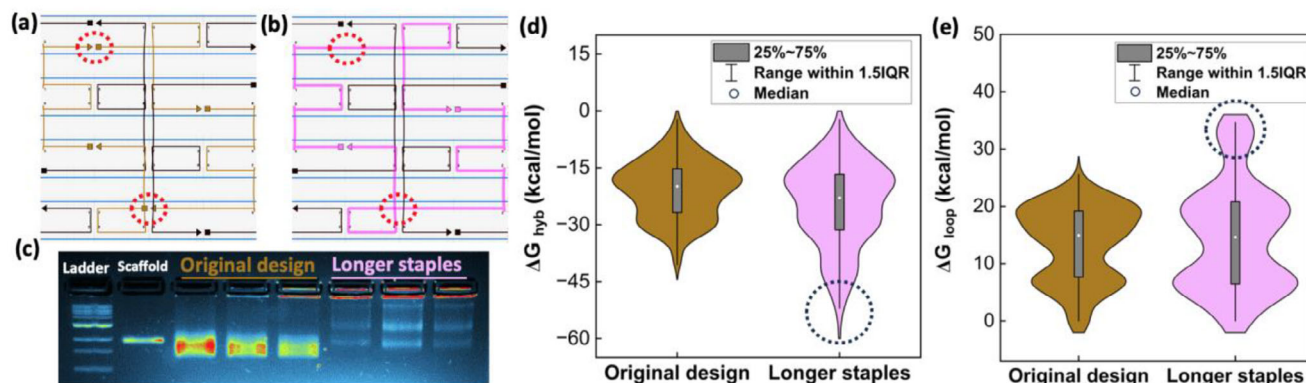


Figure 4. Effect of staple design on folding. Scaffold-staple designs for the 3 arm-linkage from caDNAo show a) before and b) after increasing the length of staples by bridging the circled nicks. Pink staples are the longer modified staples, brown are the original staples, black staples are unchanged, and blue is scaffold. c) Gel electrophoresis of both designs shows proper folding with the original design, and a significant decrease in yield for the modified design. Free energy changes for all staples show the modified staples have more favorable d) ΔG_{hyb} due to the longer binding domains, but less favorable e) ΔG_{loop} due to the increased number of crossovers per staple. The calculated total free energy change between designs is near zero, but experiments show the entropic penalty had a larger impact and entirely disrupts folding.

folding yield, where the higher entropy cost from the longer staple design deteriorated the yield. This supports the findings from Schneider et al.^[37] that non-local cooperative effects govern assembly behavior.

Thus, it is the interstrand cooperativity that plays a critical role in folding process of DNA origami.^[38] As discussed, once a critical number of staples are bound, successive staple binding is facilitated and folding continues more rapidly than predicted. This parameter has been difficult to quantify in theoretical models due to its strong dependence on structure and scaffold-staple design, leading to missing information in even the best predictive folding models.^[9] Our results show that cooperativity plays a pronounced and essential role in the assembly of complex designs, in which overcoming the nucleation barrier without the cooperative folding of staples for nucleation would be difficult. **Figure 5** shows histograms of theoretical folding temperatures of all staples in each structure, as calculated using the recent model by Aksel et al.^[9] This range is compared to the experimental T_{fold} of the structure (red dashed line), as measured through real-time fluorometry (Figure 2). Staples that folded at temperatures above their theoretical T_{fold} (the structure experimental $T_{\text{fold}} >$ theoretical staple T_{fold}) are shaded in blue. According to this analysis, at least half of the staples in most structures possess a lower theoretical T_{fold} than seen experimentally. Furthermore, the theoretical T_{fold} of the entire structure is also lower than the experimental values (Table 1). This is likely due to the difficulty of fully capturing the design-dependent cooperativity in a computational model rather than local cooperative effects. Our estimation of the effect of cooperativity is given in Table 1, calculated based on the percentage of staples that possess a theoretical T_{fold} lower than the experimental T_{fold} of the structure (i.e., the fraction of staples in the shaded blue region in Figure 5).

To facilitate the ease of estimating cooperativity following this framework, we have developed a publicly available computational tool that automates the analysis of DNA origami folding data and provides an estimation of the extent of cooperativity in a given design. This tool takes inputs of fluorescence annealing data and the theoretical thermodynamic parameters for each staple strand,

calculated using the model by Aksel et al.^[9] The script computes the experimental T_{fold} of the entire structure and each folding step, if applicable, from the fluorescence data (Figure S11, Supporting Information) and compares it to the predicted T_{fold} values of individual staples to determine the cooperativity score, as defined above. By quantifying this mismatch, the tool provides a direct, data-driven measure of cooperative folding behavior.

As Table 1 shows, cooperativity in DNA origami structures is mainly influenced by the number of scaffold crossovers, as each scaffold crossover presents an additional fold in the origami with the potential to disrupt cooperative assembly. We describe cooperativity as the facilitation of neighboring staple binding. The more

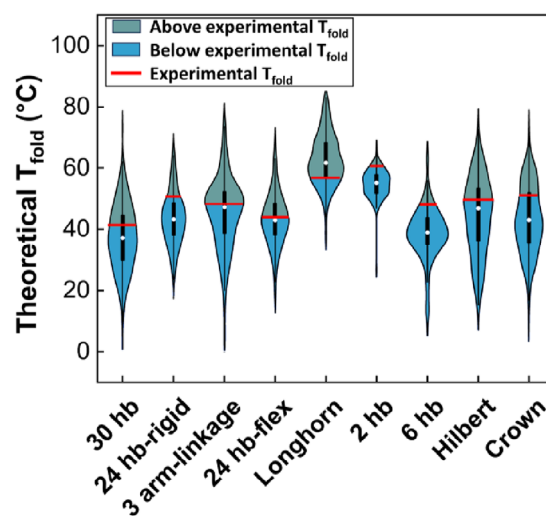


Figure 5. T_{fold} distributions for all staples in each structure calculated following Aksel et al.^[9] The blue/green boundary is the experimental T_{fold} , marked with a dashed red line, meaning the staples in the blue region have a lower theoretical T_{fold} than the experimental T_{fold} . We attribute this difference to the cooperative effect during assembly. The dark rectangles represent the 25–75% percentile of the data, the bars show the 1.5 interquartile range, and the circle is the median.

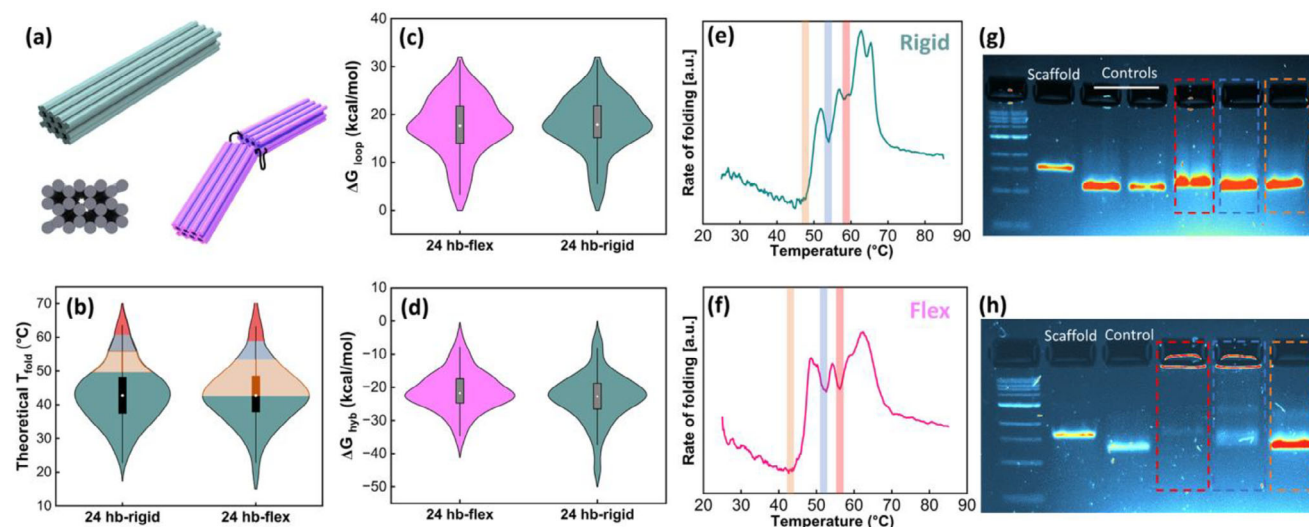


Figure 6. The impact of unbridging on cooperative assembly. a) 24 hb rigid and flex structures have identical designs except 22 staples that bridge internal scaffold crossovers are removed in the flex version. Theoretical distributions of b) T_{fold} , c) ΔG_{loop} and d) ΔG_{hyb} for all staples, calculated using code from Aksel et al.^[9] show nearly identical energies between rigid and flex versions. Real-time fluorometry curves of e) 24 hb-rigid and f) 24 hb-flex objects with three local energy minima highlighted in red, blue, and orange. Agarose gel electrophoresis analysis of g) 24 hb-rigid and h) 24 hb-flex frozen at the highlighted temperatures during the annealing ramp. Dotted boxes in gels (g–h) and shaded regions in (b) correspond to temperatures on fluorometry curves (e,f). Results show more cooperative folding in the rigid version. In (b–d), the dark rectangles represent the 25–75% percentile of the data, the bars show the 1.5 interquartile range, and the circle is the median.

folds (crossovers) in the scaffold, the more likely it is to hinder this effect. This is seen as a consistent trend across all six structures. For example, the 6 hb with 10 scaffold crossovers shows 83.5% cooperativity. Rigid 24 hb with 46 total scaffold crossovers is 72.6% cooperative, compared to the 61.7% cooperative 30 hb with 58 crossovers. At the extremes, we see the 2 hb with 98.2% cooperativity and 2 crossovers (one-fold of the scaffold) and the Longhorn with 19.2% cooperativity and 90 crossovers (≈ 45 folds). These 1D and 2D designs are expected to fold at higher temperatures compared to 3D designs. The theoretical T_{fold} (lowest folding temperature of any staple) is highest in the Longhorn, but due to the significant cooperative effects, we see the highest experimental T_{fold} in the 2 hb, as shown by the red line boundary in Figure 5 and Table 1.

Beyond the total number of scaffold crossovers, their connection via bridging staples also plays a crucial role. For rigid bundles, internal crossovers are bridged with staples, but removing these bridging staples in multi-arm designs completely disrupts cooperative folding, limiting the effect to one arm instead of the entire structure. In structures like 24 hb-flex and the 3-arm linkage, the absence of bridging staples noticeably decreases the extent of cooperativity. The 1-arm (rigid) and 2-arm (flex) versions of the 24 hb have identical scaffold designs with 46 crossovers, but the flexible version is 21.5% less cooperative. Structure design details are available in Figure S13 (Supporting Information). Hilbert and Crown structures consist of short arms with simple cross sections of 6, 8, and 12 hb with few or no embedded helices. As a result, intrastrand cooperativity and hybridization energy are more pronounced in their effect on folding. In small clusters, especially, the presence of uninterrupted 14 nt seeds enhances the overall cooperativity of the design. Despite the fact that the Crown possesses more scaffold crossovers, the higher number of 14 nt seeds (Table 1) led to a higher cooperativity compared to

the Hilbert object. Overall, fewer scaffold crossovers (especially unbridged ones) and more 14 nt scaffold-staple seeds result in a higher cooperativity in DNA origami structures, where the latter can be prioritized in structures with fewer embedded helices (simpler cross sections).

To further examine the role of cooperativity in DNA origami assembly, a 24 hb structure was designed with two variations: one rigid bundle and one flexible (flex) version with a scaffold-connected joint between 2 rigid arms (Figure 6a). Both structures share an identical scaffold-staple design, except the flex version excludes 22 central staples across internal scaffold crossovers, while the rigid version retained them (Figure S13b,c, Supporting Information), inspired by previous work.^[39] Figure 6b shows the theoretical T_{fold} for all staples. ΔG_{loop} and ΔG_{hyb} plots (Figure 6c,d) indicate that the staple free energies for both versions are nearly identical with the exception of the middle bridging staples. Although the bridging staples depict low ΔG_{hyb} (favorable), they introduce higher ΔG_{loop} (unfavorable) by adding crossovers bridging the two arms and higher ΔG_{bind} (unfavorable). The result of these subtle changes in theoretical folding behavior is negligible, with a total structure T_{fold} of 42.7 °C for the rigid 24 hb and 42.8 °C for the flexible version, as depicted in Figure 6e. However, despite this theoretical similarity, our experimental results reveal a T_{fold} of 48.7 and 43.0 °C for the rigid and flexible 24 hb structures, respectively (Figure 6e,f). This tells us that the degree of cooperativity is lower in the flex version than in the rigid version, as shown with full thermal ramps. Connecting the two halves in the rigid version extends the cooperative effect throughout the entire structure, facilitating formation of the other arm, compared to the 2-arm flexible version, which limits the cooperative enhancement to a single arm.

To investigate these differences more thoroughly, structures were shock-frozen at the temperatures highlighted in Figure 6e,f,

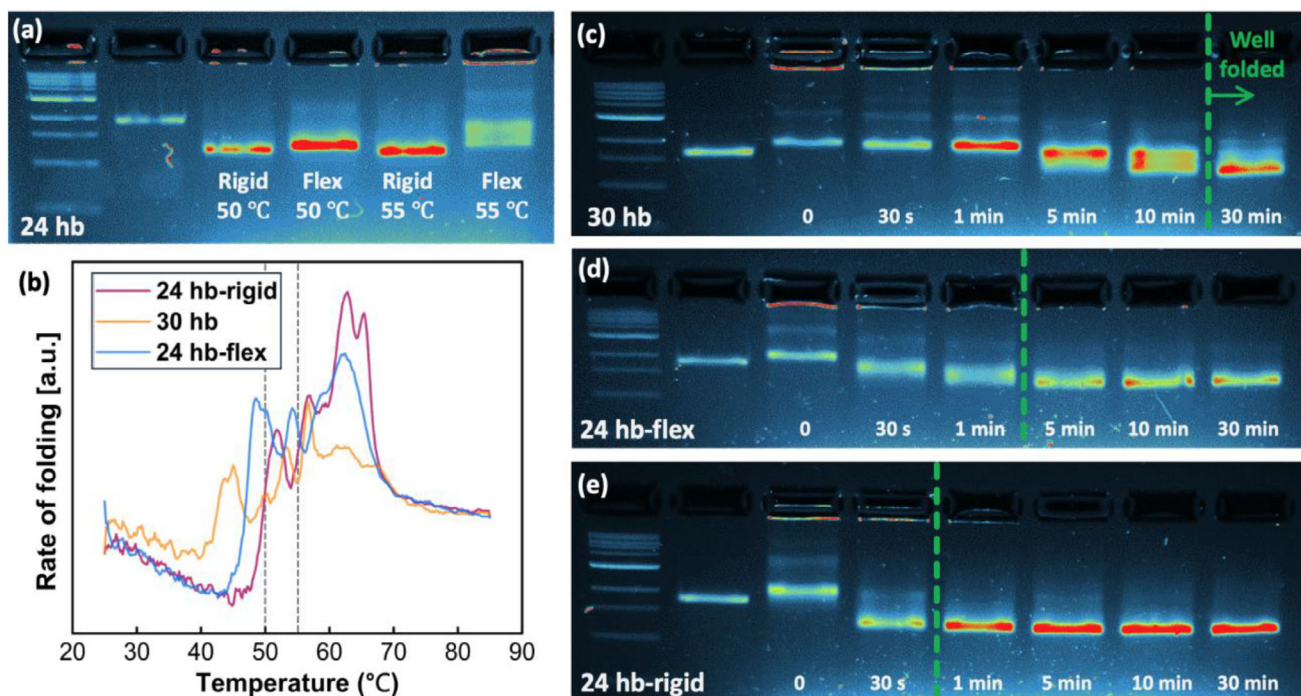


Figure 7. Kinetics of constant-temperature folding at T_{fold} . a) Gel electrophoresis analysis of 24 hb rigid and flex objects folding at constant 50 and 55 °C for 30 min. b) Real-time fluorimetry results for the three structures studied here. Gel electrophoresis was used to evaluate the folding of c) 30 hb, d) 24 hb-flex and e) 24 hb-rigid objects after heating at their T_{fold} , (41, 43, and 47 °C, respectively) for the times indicated. Proper folding is indicated by the green dashed line with well-folded structures to the right of the line. The first and second lanes of all gel images are ladder and 7249-scaffold, respectively.

where a dip in the rate of folding is seen. Gel electrophoresis analysis in Figure 6g,h shows the extent of folding at each temperature, showing near complete folding for the rigid, but almost nothing resembling folded structures at intermediate temperatures for the flex design. Control bands show properly folded structures. For comparison, theoretical T_{fold} is shown in Figure 6b with each temperature range highlighted, again showing a large discrepancy between theoretical and experimental T_{fold} due to the role of cooperativity. This is apparent with the intermediate folding analysis and higher T_{fold} of the rigid structure, indicating a more robust and faster assembly with the majority of folding occurring during the first ≈ 60 °C folding step, well above the theoretical T_{fold} .

As reported previously,^[40–42] DNA origami structures can be folded under isothermal conditions at temperatures close to their T_{fold} . To measure their robustness, both versions of the 24 hb were heated at constant temperatures of 50 and 55 °C for 30 min, intermediate ranges above their T_{fold} , and their gel electrophoresis analysis is shown in Figure 7a. Whereas the rigid structure formed at both temperatures, the flex design only assembled at 50 °C, suggesting a less robust and less cooperative design. Based on the annealing profiles (Figure 7b), at 55 °C the rigid structure was farther along in the folding process than the flex structure and folding was likely able to complete with enough time due to cooperative effects, even at a temperature higher than T_{fold} . Furthermore, to study their constant temperature folding kinetics, the 30 hb, 24 hb-rigid, and 24 hb-flex structures were heated at their T_{fold} (Table 1) for 0, 0.5, 1, 5, 10,

and 30 min (gel electrophoresis images shown in Figure 7c–e). The required time to properly assemble each structure with near 100% yield was 30, 5, and 1 min for the 30 hb, 24 hb-flex, and 24 hb-rigid, respectively. Easier folding is apparent for a simpler cross section with fewer embedded helices (24 hb vs 30 hb) and for a continuous structure compared to a jointed 2-arm bundle. Thus, folding kinetics appear to be influenced more by structure design than cooperativity. Folding yield and migration distance metrics for these gels are provided in Table S2 (Supporting Information).

DNA origami structures with more complicated designs (e.g., more crossovers per staple and higher ΔG_{loop}) as well as structures with higher nucleation barriers that rely more on cooperativity, require longer times for nucleation and complete assembly when annealed at constant temperatures.^[24,8] Consequently, in the case of a complex design, the probability of rapid and successful pairing of each staple may be lower or slower, as the staples will require lower temperatures and/or longer times to anneal due to the larger nucleation barrier, hence, diminishing or delaying the “helper” ability for subsequent staple pairing. This is apparent with the higher nucleation barrier in the 24 hb flex and 30 hb structures that demands a stronger cooperative effect and results in a longer annealing time necessary for efficient folding. The cooperativity influence is terminated when staples do not maintain coaxial stacking to reduce entropy cost for neighboring staples, such as the disjointed 24 hb flex design. This leads to lower T_{fold} and slower folding.

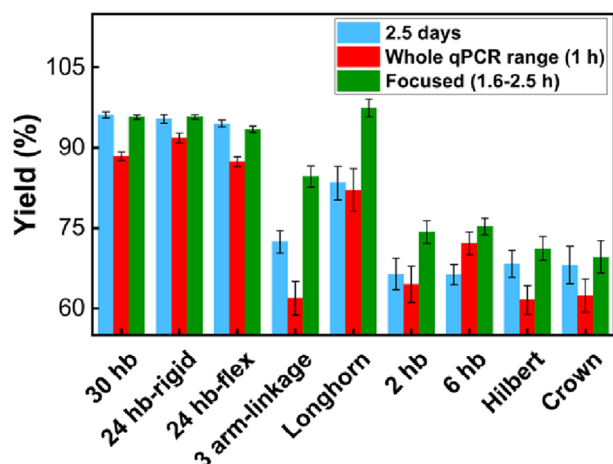


Figure 8. Folding yield analysis of all structures after thermal annealing ramps of 2.5 days, the whole real-time fluorometry range used in Figure 2 (1 h total), and new focused thermal ramps covering only the temperatures in the folding range of the structure (Figure 2j) at 5 min °C⁻¹. The corresponding gel electrophoresis images are shown in Figure S12 (Supporting Information).

2.4. Strategies to Enhance Folding Yield

From our results presented here and the cumulative literature we have discussed, we identify many factors that affect the folding yield of DNA origami. Scaffold-staple design and thermal ramp protocols have a more significant effect once the salt concentration is optimized.^[35,43] Regarding the scaffold-staple design, the energy changes of each staple during binding events should be considered, and the trade-off between the energy of hybridization and loop formation should be optimized, where the latter has shown a more pronounced impact. Concerning the thermal ramp, certain temperatures should be focused on or completely avoided to achieve the maximum yield. Real-time fluorometry is a valuable technique that can be utilized to fulfill this goal by determining the temperatures at which folding is initiated and completed. This is unique for every design and sensitive to even a few staple changes. These considerations are also crucial for multicomponent structures, where the sequence of folding sections matters.^[42,44] Additionally, through a folding and unfolding process, this technique can assess whether the thermal ramp used resulted in the formation of the intended DNA origami structure or caused agglomeration and undesirable base pairing events, without the need for gel electrophoresis or TEM (Figures S4–S6, Supporting Information). Finally, in some cases, heating for long enough above the temperature where folding starts can reduce the folding yield or even prevent it due to increased undesirable aggregation (Figures S4–S6, Supporting Information).

By focusing on the folding temperature range, not only can the yield be improved, but the folding time can be noticeably lowered. **Figure 8** compares the folding yield for thermal ramps of 2.5 days, a 1 h constant ramp of the full temperature range (same as Figure 2), and a ramp focused on the folding temperature range (1.6–2.5 h). The associated gel electrophoresis images are shown in Figure S12 (Supporting Information). For the focused ramp, structures were annealed only in their folding range, decreasing

the temperature by 0.1 °C every 30 s. The 2.5-day and 1-h ramps covered the full temperature range from 65 to 25 °C. The results in Figure 8 show that the focused ramp achieved equal or better folding yield than the 2.5 days or rapid full-range ramps for all structures by avoiding aggregation and kinetic traps at unnecessary temperature ranges. For instance, the folding yield of the Longhorn and 3-arm linkage were enhanced by ≈17%, compared to the 2.5-day ramp, and the latter increased yield by 22.7% compared to the rapid 1-h ramp. Performing one real-time fluorometry folding experiment to determine the optimal folding range, then using this strategy for enhanced rapid folding can save significant time while maximizing folding yield.

We have established two important lessons to enhance DNA origami folding yield: 1) in the design phase, ΔG_{loop} is of paramount importance and should be minimized while preserving the design integrity, and 2) real-time fluorometry is a valuable tool that gives insight into the folding temperature range of the intended design, and thus, an optimized thermal ramp can be designed to focus on this range. In addition, to develop a multi-component device, cooperativity and nucleation barriers can be tailored for directed assembly.

3. Conclusion

This study examined the folding behavior of DNA origami, revealing the impact of design parameters on free energy changes, cooperativity, folding yield, and folding kinetics. Real-time fluorometry analysis showed that structures with more intricate designs (e.g., embedded helices or jointed arms) exhibited lower T_{fold} and multiple folding steps. Energy analysis of 3-arm linkages indicated that loop formation energy (entropy) plays a larger role than hybridization energy, with long staples hindering folding due to increased entropy cost, despite more favorable binding domains. We suggest a higher yield can be obtained by increasing the length of staples to prioritize more 14+ nt scaffold-staple base pairs, as long as the number of staple crossovers is less than four. We show that cooperativity in DNA origami folding is strongly influenced by the number of scaffold crossovers and their potential connections via bridging staples. In rigid structures, fewer scaffold crossovers enhanced cooperativity, ranging from 98% cooperative with two crossovers to 19% cooperative with 90 crossovers. Removing bridging staples between internal scaffold crossovers, which creates multi-arm structures, was shown to decrease cooperativity by over 20% in the 24 hb, while lowering T_{fold} and slowing assembly. A high number of 14 nt scaffold-staple base pairs in the Crown structure suggested the enhanced intrastrand cooperativity, leading to the improved overall cooperativity. Quantifying the control of design-dependent cooperativity and its impact on folding have been missing in previous experimental and computational studies. With the newly introduced tool, one can calculate the role of cooperativity from fluorescence data and theoretical folding energies. Building from the recent computational models from Aksel et al.,^[9] our experimental results can be used to improve the yield and assembly time of complex DNA origami nanostructures. This is demonstrated with focused annealing ramps based on new insights from fluorometry results that enhance yield up to 17% while cutting the required assembly time to just 1–2 h. This impact was especially noticeable for complex structures like the 3-arm linkage

by focusing on the necessary temperatures and preventing aggregation at unnecessary temperatures. Optimizing free energy contributions and the extent of cooperativity in the design phase are essential for overcoming cost, yield, and efficiency limitations in large and complex DNA origami assemblies.

4. Experimental Section

DNA Origami Designs and Fabrication: DNA origami computer-aided design software caDNAno^[45] was employed to design all structures (Figure S9, Supporting Information). Finite element modeling tool CanDo^[46,47] was also used to analyze and inform designs. In addition, a recently introduced computational tool^[9] was used to calculate the energy profiles. The structures were fabricated using 7249-nucleotide (30 hb, 24 hb-rigid, 24 hb-flex, 2 hb, 6 hb) and 8064-nucleotide (3 arm linkage, Longhorn logo, Hilbert, Crown) scaffolds (Tibit Co., Germany and Bayou Biolabs) and engineered staple oligos ordered were from Integrated DNA Technology. For the self-assembly process of all studied structures, 10 nM scaffold with the associated 100 nM staples were mixed in folding tubes, including 50 mM Tris, 10 mM EDTA and 18 mM MgCl₂, similar to the process employed by Castro et al.^[47] This was followed by a thermal annealing ramp where the temperature decreased from 85 to 25 °C by −0.2 °C and heating at each temperature for 13 s. In addition, for the focused thermal ramps, structures were heated in their folding temperature range where data was obtained from real-time fluorometry results. Here, the temperature was decreased by −0.1 °C and held at each temperature for 30 s. Also, for constant temperature annealing, structures were held at the corresponding temperatures for the studied time points. For comparison, thermal annealing was conducted over 2.5 days, starting at 65 °C and cooling to room temperature with a gradual decrease of 1 °C per step. The sample was held at each temperature for 1 h until reaching 62 °C, 2 h until 59 °C, 3 h until 46 °C, 1 h until 40 °C, and finally 30 min until 25 °C.

Structure Characterization Using Gel Electrophoresis and TEM: To confirm the folding efficiency, the reaction products were mixed with a homemade loading dye electrophoresed on gels containing 2% agarose in 0.5x TBE with 11 mM MgCl₂ and 1 μM SYBR safe for 1.30 h at 70 V in an ice bath. Analytik Jena UVP GelStudio imager (Jena, Germany) was used to capture and process gel images. Camera exposure was adjusted such that the most intense band was below the saturation threshold, preventing signal clipping and allowing for accurate relative comparison of band intensities. After capturing gel images, Analytik Jena VisionWorks software was used to apply a heatmap filter based on intensity for ease of visualization. The fast-leading band for structures in different wells were cut and purified utilizing freeze and squeeze DNA gel extraction spin columns (Bio-rad, Hercules, CA). For TEM imaging, 5 μL of purified objects were deposited and incubated for 1 min on glow-discharged formvar-supported carbon-coated TEM grids (Electron Microscopy Sciences, Hatfield, PA) and stained utilizing 1% uranyl acetate. Images were taken at 80 kV acceleration voltage with a FEI Tecnai G2 Spirit TEM.

Real-Time Fluorometry: DNA origami objects were studied using a real-time PCR system (Bio-rad, CFX Opus 96) with the thermal ramp discussed in the fabrication section. In this experiment, the fluorescence intensity changes for the DNA origami structures were subtracted from their corresponding staple-only samples, and the derivative of the subtracted values that was denoted as the “rate of folding” was plotted against temperature. Similar to the folding studies, unfolding analysis was performed once the structures were fully formed by heating from 25 to 85 °C. The heating/cooling ramp was optimized by trying different ramps (Figures S4–S6, Supporting Information) and kept consistent for all studied structures to exclude its influence on the assembly process. This ramp was implemented by increasing/decreasing the temperature by 0.2 °C and heating/cooling at each temperature for 13 s.

Statistical Analysis: All statistical analyses and data visualization were performed using OriginPro 2025. Gel electrophoresis images were processed with VisionWorks software to calculate yield. For the DNA origami yield experiment, data were presented as mean ± standard deviation (SD).

In violin plots, the dark rectangles represent the 25–75% percentile of the data, the bars show the 1.5 interquartile range, and the circle was the median.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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